

Conditional diffusion emulator of Veros ACC equilibrium states: first end-to-end results

2 October 2026

Summary

- Goal: generate the post-spin-up (T, S) state of the Veros ACC channel from the EKE-closure coefficients (c_k, c_ϵ) , skipping the spin-up.
- Pipeline: DINO-Fusion diffusion code adapted to a parameter ensemble; one split runs end to end in under one A100 hour.
- Interpolation (scattered points, a band of three rows, the 5×5 and 7×7 centre blocks held out): hold-out RMSE 0.061 / 0.076 / 0.061 / 0.064 K against 0.037 / 0.116 / 0.077 / 0.114 K for the nearest training run and 0.155–0.197 K for the mean state. The model’s error does not depend on the size of the gap; the baselines’ does, so the model wins as soon as the gap exceeds one grid step.
- Extrapolation (the 51 or 75 outer runs held out, training on the centre; the top row): 0.110 / 0.138 / 0.141 K against 0.131 / 0.155 / 0.170 K, medians 0.061 / 0.064. The mean is set by the warm corner, where the model is cold-biased (corner run 0.77 and 1.10 K when it lies two and three cells from the training set); elsewhere 0.04–0.13 K. A Wasserstein metric on the stratification profiles gives the same ranking.
- Four training runs only, one step in from each corner: the model still beats the nearest run on average (0.154 vs 0.179 K) and in W_1 (0.106 vs 0.156 K) because it interpolates the warm-side trend the baseline cannot, but its error is now 0.1–0.45 K over most of the grid.
- Physical miss: temperature inversions are generated at the rate of the weakly eddying runs in every regime.

1 Data

- 100 Veros ACC runs on a 10×10 log-spaced grid, $c_k \in [0.0125, 0.8]$, $c_\epsilon \in [0.0875, 5.6]$; 100 years, 30-day output; interior grid $30 \times 42 \times 15$ (x, y, z).
- Kept: T and S , last 20 years (241 snapshots per run). $S = 35$ psu everywhere in every run.
- Nearly deterministic end states: within-window std 0.01 K above 1400 m vs 0.05 K between grid neighbours; the bottom level still drifts by 0.45 K over the window.
- One effective parameter: $\log c_k - 1.05 \log c_\epsilon$ explains 99 % of the variance of the domain-mean T ; the low- c_k/c_ϵ part of the grid saturates.
- Per-level distribution of T (Figure 1): above 400 m a mixture of restoring values and interior, with σ_z set by the meridional gradient, so $\mu_z \pm 3\sigma_z$ is wider than the level and the normalised values stay within $[-0.85, 0.6]$; below 1400 m a warm tail to $x' = +2.7$. Run-to-run signal σ_z^{runs} : 0.7 % of $3\sigma_z$ at the surface, 26 % at the bottom. The narrow peaks are the zonally uniform southern rows $y = 0\text{--}6$, the restoring zone: one peak per row, at the same value in every run to within 0.04–0.18 K (Figure 2).

Distribution of T per level, all 100 runs $\times$ 241 snapshots, water cells (surface at the top). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

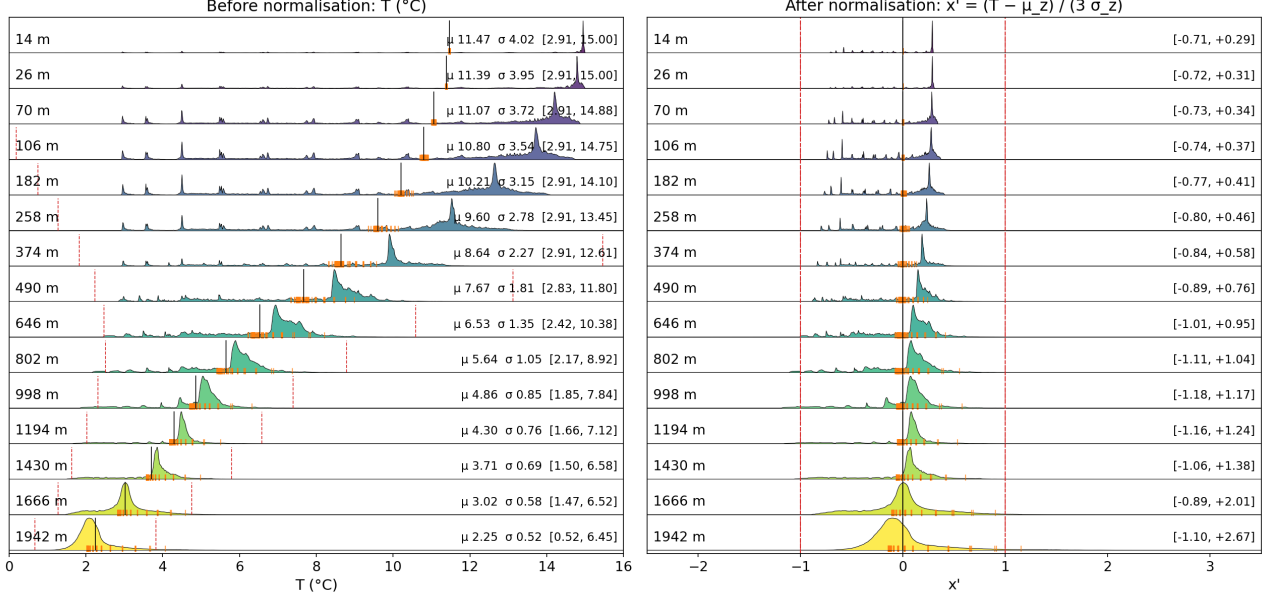


Figure 1: Distribution of T per level over all 100 runs and 241 snapshots (water cells), before (left) and after (right) the per-level normalisation of Eq. (1). Black: μ_z ; red dashed: $\mu_z \pm 3\sigma_z$; orange ticks: the 100 per-run level means.

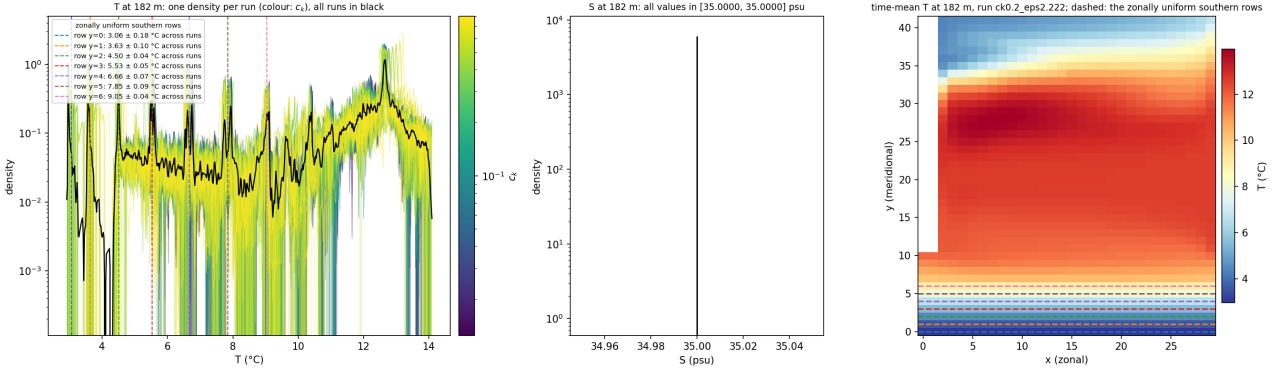


Figure 2: The level nearest 150 m (182m), last 20 years, water cells. Left: one density of T per run (colour: c_k ; log scale), all runs in black; dashed: the zonal-mean T of the southern rows $y = 0-6$, each with its spread across runs. Middle: S , exactly 35psu everywhere. Right: time-mean T of one run at that level with the same rows marked. Every narrow peak of the distribution is one southern row.

2 Method

- Fields $\rightarrow$ 30 channels ($T_{1..15}, S_{1..15}$), normalised per level, land $\rightarrow$ 0, zero-padded $42 \times 30 \rightarrow 48 \times 32$:

$$x'_z = \frac{x_z - \mu_z}{3 \max(\sigma_z, 0.05)}. \quad (1)$$

μ_z, σ_z are computed once on all 100 runs and held fixed for every split (a deliberate, mild leakage of 30 scaling constants; no held-out state enters training).

- Denoiser: UNet2DModel (64, 64, 128, 128), 6.9 M parameters. The condition c , the standardised pair ($\log c_k, \log c_e$), enters through an MLP added to the timestep embedding.
- DDPM, ϵ -prediction, 1000 steps, squared-cosine schedule, EMA:

$$\mathcal{L}(\theta) = \mathbb{E}_{t, x_0, \epsilon} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t, c)\|^2. \quad (2)$$

20 000 steps, batch 32, AdamW 10^{-4} with cosine decay; 22 min on one A100.

- Sampling: 1000-step DDPM loop with EMA weights; the predicted clean state is clipped at $|x'| \leq 1$; after every step the land and padding cells are reset to zero at the step’s noise level, $\sqrt{1 - \bar{\alpha}_t} z$, as in training. 32 samples per condition.

Differences from the DINO-Fusion code.

- Conditioning: DINO’s UNet is unconditional and is steered at sampling time by constraints; here the UNet receives $(\log c_k, \log c_\epsilon)$ through the timestep embedding, and the data are an ensemble of 100 runs split by run instead of one trajectory.
- Sampling: DINO’s BorderZeroConstraint resets land and padding to exact zeros after every step, which the network never sees in training, where those cells carry noise like every other. Here they are reset to zeros at the step’s noise level, $\sqrt{1 - \bar{\alpha}_t} z$, exactly as in training. DINO’s stratification constraints are not ported yet.
- Everything else is kept: transform chain, 3-std normalisation, DDPM settings, EMA.

3 Evaluation

- Truth: 20-year time-mean state of each held-out run; metrics on water cells.
- Baselines: mean training state; nearest training run in standardised log-parameter space; average of the axis-neighbour training runs on the grid.
- Eight hold-out splits, same data file and statistics, different training sets. Interpolation: scattered (10 interior points), band of three rows ($c_k = 0.126, 0.2, 0.3175$; 30 runs; training rows a factor 6.3 apart), centre 5×5 (c_k 0.05–0.32, c_ϵ 0.35–2.2; 25 runs) and centre 7×7 (c_k 0.0315–0.504, c_ϵ 0.22–3.5; 49 runs; training on the outer rows and columns only). Extrapolation: outer 51 (the complement of the 7×7 block, training on its centre), outer 75 (the complement of the 5×5 block, training on 25 runs), top row ($c_k = 0.8$; 10 runs). Four runs: training on $c_k \in \{0.0198, 0.504\} \times c_\epsilon \in \{0.139, 3.53\}$, one step in from each corner; 96 held out.
- Distributional metric: each state is reduced to its profile of horizontal means over water cells, $\bar{T}_z$; per level, the Wasserstein-1 distance between the 32 generated values and the 241 true snapshots of the window, through the quantile functions,

$$W_1^{(z)} = \int_0^1 |F_{g,z}^{-1}(q) - F_{t,z}^{-1}(q)| dq, \quad W_1 = \sum_z \frac{\Delta z_z}{H} W_1^{(z)}, \quad (3)$$

with Δz_z the level thicknesses ($H = 2080$ m). All 241 snapshots are used: the quantile form accepts unequal sizes and a subsample of 32 would only add noise. A point prediction x (the baselines) gives $W_1^{(z)} = \frac{1}{241} \sum_i |x_z - \bar{T}_{z,i}|$. W_1 rewards a correct spread and does not reward collapse to the mean; the RMSE keeps the horizontal structure.

split	runs	RMSE (K)				bias (K)	W ₁ (K)			inversions (%) generated / truth
		mean of 32	sample	nearest	mean state		diffusion	nearest	mean state	
scattered	10	0.061	0.147	0.037	0.155	+0.04	0.066	0.044	0.136	10.2 / 7.7
band of three rows	30	0.076	0.160	0.116	0.197	+0.02	0.075	0.095	0.160	10.9 / 1.3
centre 5×5	25	0.061	0.153	0.077	0.157	+0.04	0.070	0.069	0.133	10.4 / 3.6
centre 7×7	49	0.064	0.153	0.114	0.197	+0.03	0.073	0.095	0.166	10.4 / 4.2
outer 51	51	0.110	0.185	0.131	0.258	−0.00	0.100	0.117	0.223	11.0 / 6.6
outer 75	75	0.138	0.213	0.155	0.220	−0.02	0.118	0.137	0.189	10.6 / 6.1
top row	10	0.141	0.208	0.170	0.470	−0.02	0.132	0.142	0.420	12.3 / 0.1
four runs	96	0.154	0.214	0.179	0.254	+0.05	0.106	0.156	0.208	10.5 / 5.4

Table 1: Hold-out metrics per split, mean over the held-out runs, water cells, 32 samples per run. Outer 51 and outer 75: the complements of the 7×7 and 5×5 blocks, training on the block; four runs: training on one run one step in from each corner. Bias: domain mean of the ensemble mean. The neighbour-average baseline equals the nearest run except in the scattered split (RMSE 0.020 K, W_1 0.036 K); the ensemble spread is 0.12–0.13 K against a true within-window spread of 0.03 K; the salinity error is 0.012–0.017 psu in every split.

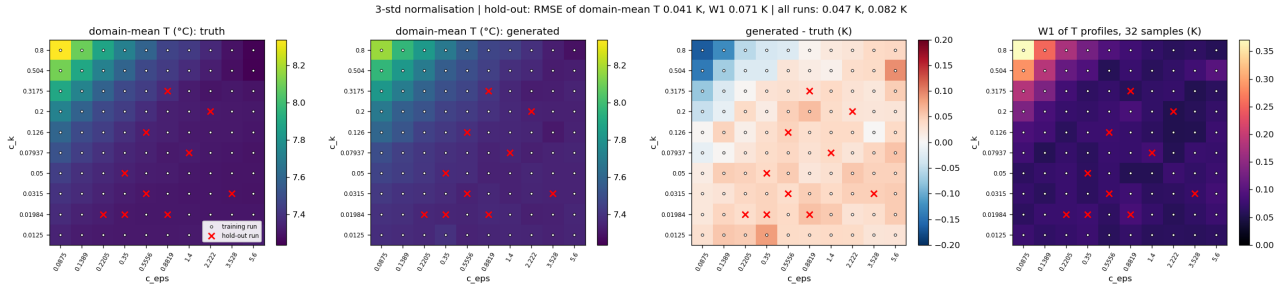
4 Findings

- **Interpolation: the model’s error is gap-independent, the baselines’ is not.** Band: beats the nearest run at every level below 26 m and in all three rows (0.063 / 0.074 / 0.091 K vs 0.063 / 0.132 / 0.152 K for $c_k = 0.126$ / 0.2 / 0.3175). Centre 5×5 and 7×7 : the same error, 0.061 and 0.064 K, flat over the block (0.04–0.12 K), while the nearest run goes from 0.077 to 0.114 K and reaches 0.18 K at the centre of either block, where the model stays at 0.05 K. In every split the losses are the runs of the flat regime (low c_k , high c_e), where a copied neighbour is already within 0.05 K; the scattered split, whose neighbours are all one step away, is the one where the nearest run wins overall (Figures 3–6).
- **Extrapolation: the warm corner sets the mean, the rest behaves like interpolation.** Outer 51 / outer 75 / top row: mean 0.110 / 0.138 / 0.141 K, median 0.061 / 0.064 / 0.11 K. The corner run ($c_k = 0.8$, $c_e = 0.0875$) errs by 0.77, 1.10 and 0.41 K with a cold bias (−0.43, −0.68, −0.19 K): the model does not carry the warming beyond the last training row, and the error grows with the distance to it (two, three and one cells). Away from the corner the outer 75 sit at 0.089 K against 0.095 K for the nearest run. The model beats the nearest run from about 150 m to the bottom and at 24 / 51, 32 / 75 and 7 / 10 runs; the nearest run is worse still at the corner (0.92 and 1.24 K) (Figures 7–9).
- **Four training runs.** Mean 0.154 vs 0.179 K for the nearest run and 0.254 K for the mean state, W_1 0.106 vs 0.156 K, but median 0.120 vs 0.070 K and 31 wins of 96. The model produces a smooth field over the grid, 0.05–0.16 K in the flat regime and 0.2–0.45 K on the warm side with a warm bias of up to 0.27 K between the two c_k rows it was trained on, and 0.70 K at the corner (cold). The nearest run is far better where the state hardly changes (0.01–0.07 K) and far worse where it does (0.5–0.85 K in the warm half of the interior), so the two average out differently: inside the training rectangle 0.159 vs 0.201 K, outside 0.146 vs 0.143 K. Below 250 m the model wins at every level (Figure 10).
- **Depth structure.** 0.03–0.08 K at every level in the interpolation splits; the bottom levels are no longer special (0.05–0.07 K) except where the warm corner is held out (top row 0.29 K, outer 51 0.17 K, outer 75 0.19 K).
- **Why the surface is worst in kelvin.** The per-level scale $3\sigma_z$ is 12 K at the surface and 1.6 K at the bottom, and the sampler’s precision is roughly uniform in normalised units: the surface RMSE is 0.6 % of its scale (0.073 K, scattered), against 1 % at 650 m and 4.5 % at the bottom. The run-to-run surface signal is 0.7 % of the scale (0.085 K), the same size, so the model cannot tell the runs apart there and matches the mean state (0.076 vs 0.073 K); the nearest run wins (0.026 K) only because neighbouring runs are nearly identical at the surface. What remains is a per-sample offset of the surface mean: W_1 on the horizontal-mean profile is 0.12 K at 14 m, above the 0.07 K cell RMSE of the ensemble mean, so each sample carries a nearly uniform shift of about 0.1 K (0.01 normalised) that the average of 32 mostly removes.
- **The clip trades regularisation for range.** Same models, sampling only, scattered / top row: clip at $|x'| \leq 1$ gives 0.061 / 0.141 K, at 1.5 it gives 0.061 / 0.108 K, at 3 it gives 0.091 / 0.123 K; W_1 0.066 / 0.132, 0.086 / 0.117, 0.147 / 0.154 K. Widening to 1.5 costs nothing in interpolation RMSE, widens the sample spread (W_1), and frees the top-row corner, where 12 % of the true bottom values lie above $\mu_z + 3\sigma_z$ and cannot be generated under the clip at 1; at 3 the regularisation is lost (0.091 and 0.147 K in the scattered split). The corner run is the largest error of the top-row and outer splits under the clip at 1.
- **Inversions.** 10–12 % of interfaces with $\partial_z T < 0$ in every regime; truth 7.7 % (scattered), 6.6 and 6.1 % (outer 51 and 75), 4.2 and 3.6 % (centre 7×7 and 5×5), 1.3 % (band), 0.1 % (top). Not learned.
- **W_1 agrees with the RMSE ranking.** Training grid points score like held-out ones (0.083 vs 0.071 K, scattered); from 490 m down the samples beat the point baselines in the band and top splits (0.058 vs 0.115 K at 650 m, band), which pay for the spread they cannot represent.
- **Plumbing.** S within 0.02 psu of 35 with no constraint; land exact; ensemble spread $4\times$ the true within-window spread (Figures 11–17).

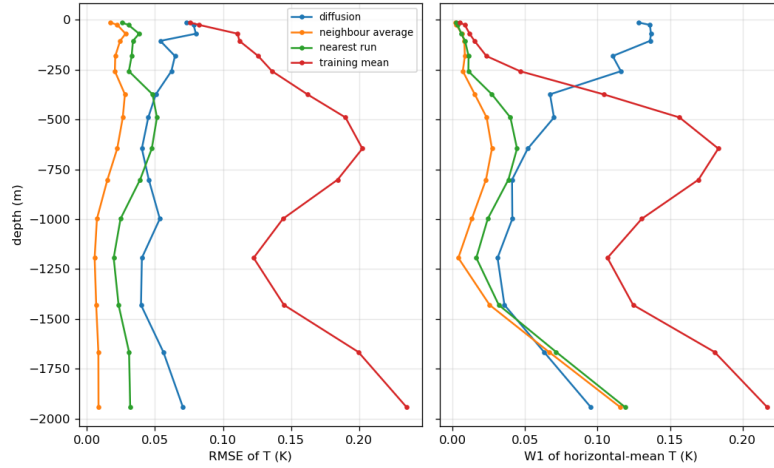
5 Next steps

- Stratification constraint at sampling time (DINO’s isotonic projection on T): targets the inversions.
- Top-row corner: the clip’s range limit is the dominant extrapolation error; a clip at 1.5 recovers 0.03 K there at no interpolation cost (sampling only), a clip that follows the conditioning is the next test.
- Stronger conditioning: classifier-free guidance; the year as a third condition (bottom drift of 0.45 K inside the window). A second training seed to size the seed noise.
- Evaluate against snapshots as well as the time mean (nearest-snapshot RMSE, spread-skill).
- Then the same pipeline on a global Veros configuration (active S , seasonal cycle, multi-century spin-up).

Reproducibility. Branch `miniveros` of the DINO-Fusion repository, package `miniveros/`. Training at `2b542aa` (338997d for the 7×7 block, `61d0dd2` for the outer 51, `8f157d5` for the 5×5 pair, `812ff96` for the four-run split); sampling and evaluation at `bfee32f`. Extraction 2.5 min (8 cores), training 17–22 min (one A100), generation of the hold-out and grid sets plus evaluation 14–18 min.

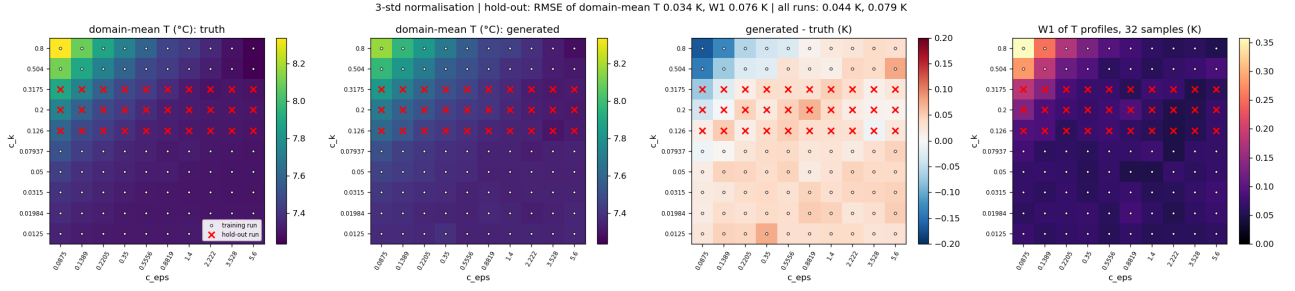


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

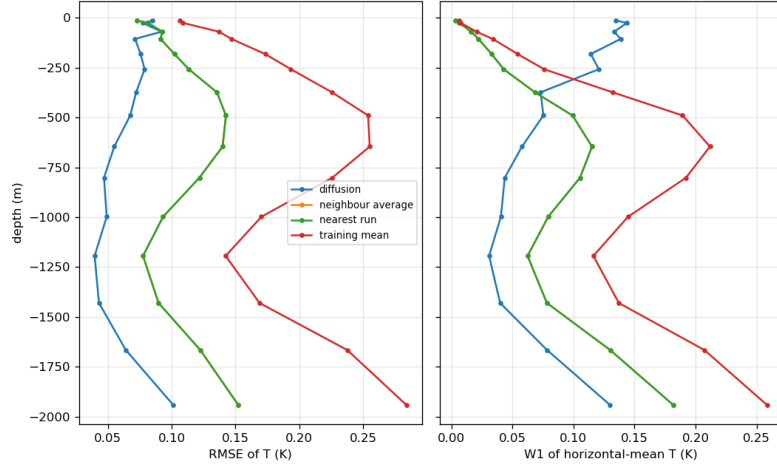


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 3: Scattered split (10 interior held-out runs), 32 samples per run.

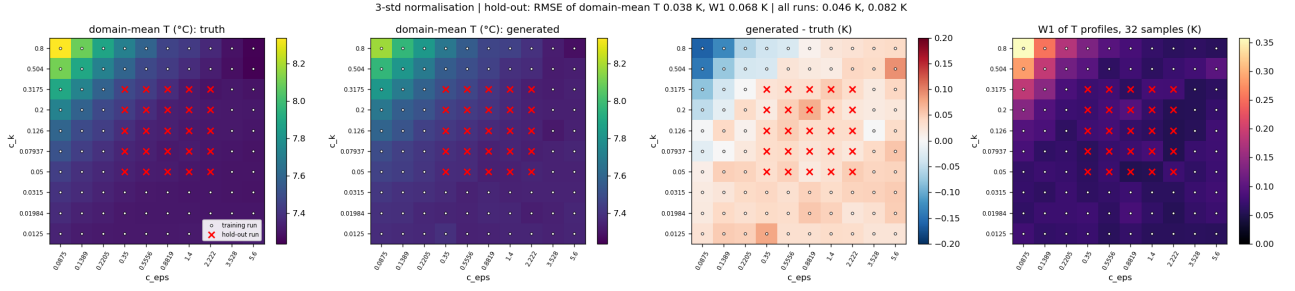


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

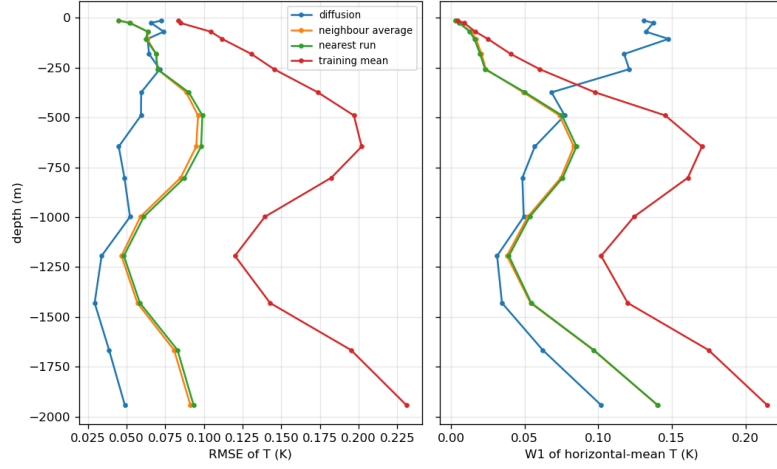


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 4: Band of three rows ($c_k = 0.126, 0.2, 0.3175$ held out), 32 samples per run.

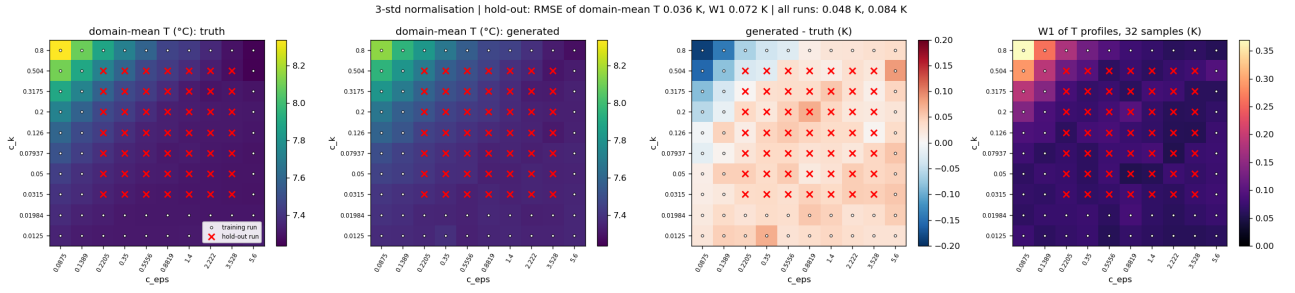


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

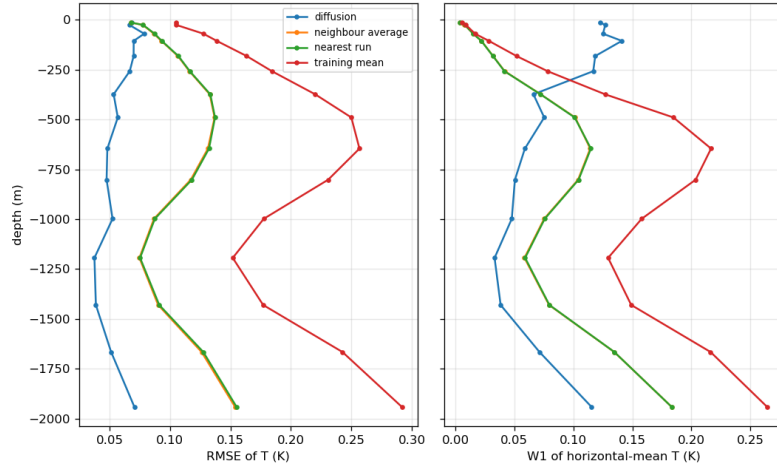


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 5: Centre 5×5 (c_k 0.05–0.32, c_ϵ 0.35–2.2 held out, 25 runs), 32 samples per run.

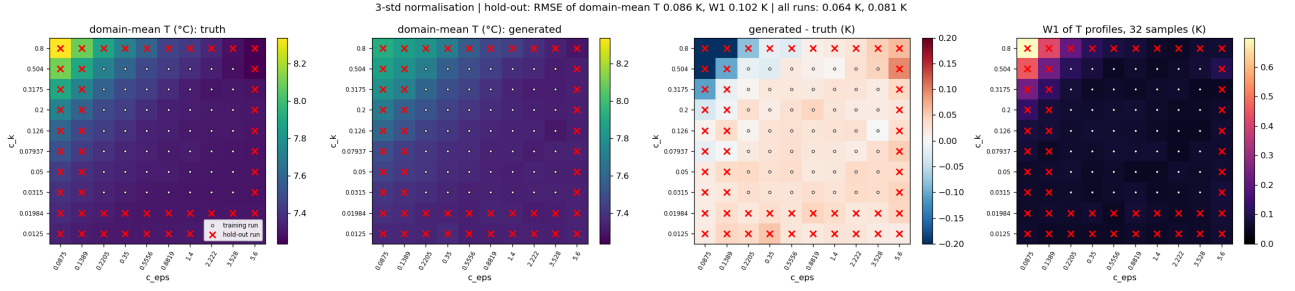


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

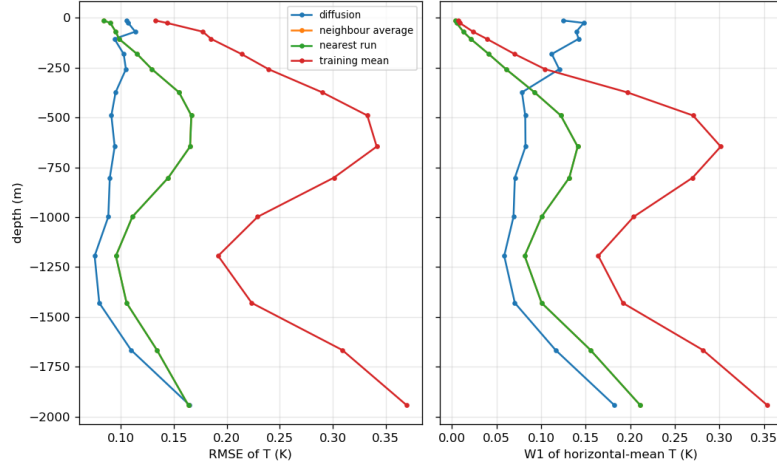


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 6: Centre 7×7 (c_k 0.0315–0.504, c_ϵ 0.22–3.5 held out, 49 runs; training on the outer rows and columns), 32 samples per run.

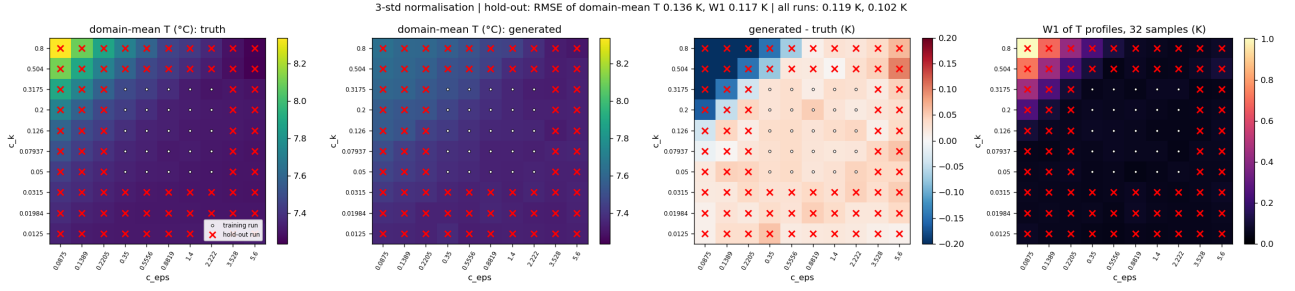


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

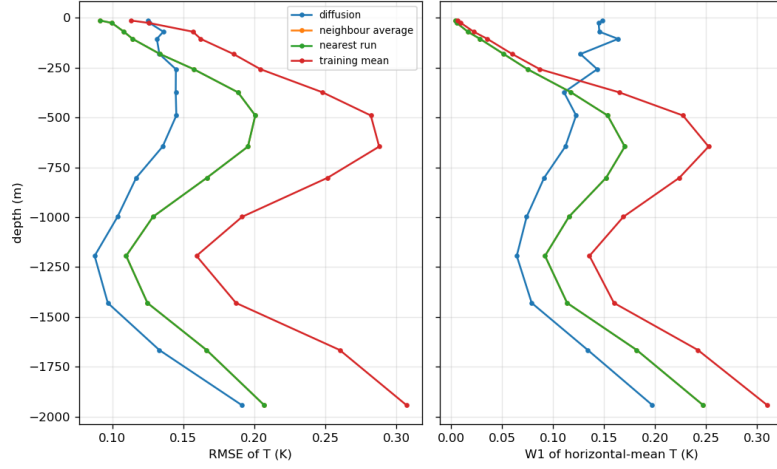


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 7: Outer 51 (the complement of the 7×7 block held out; training on its centre), 32 samples per run.

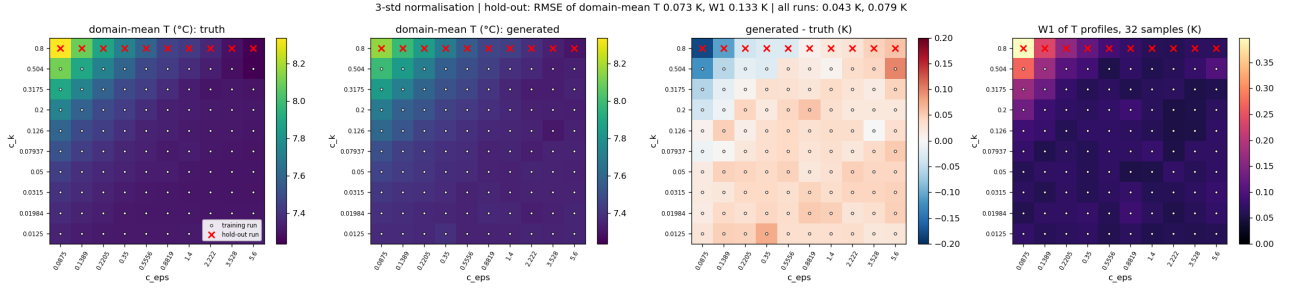


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

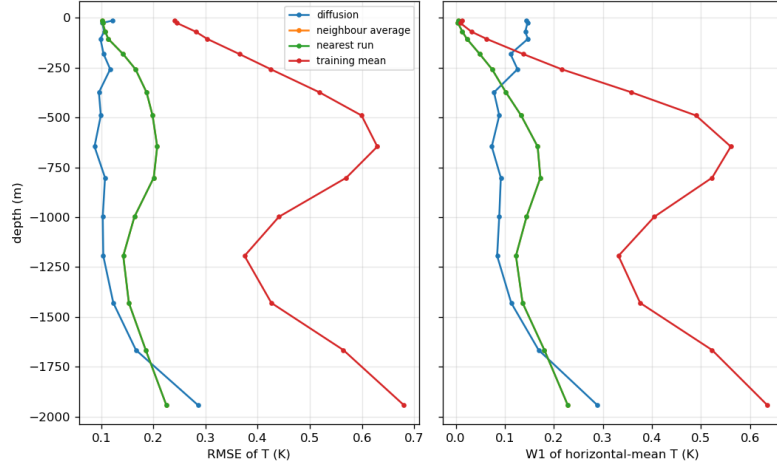


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 8: Outer 75 (the complement of the 5×5 block held out; training on 25 runs), 32 samples per run.

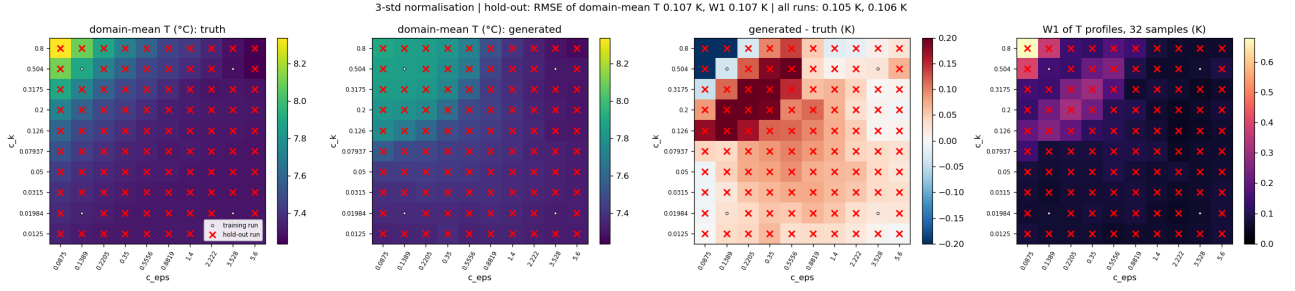


(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.

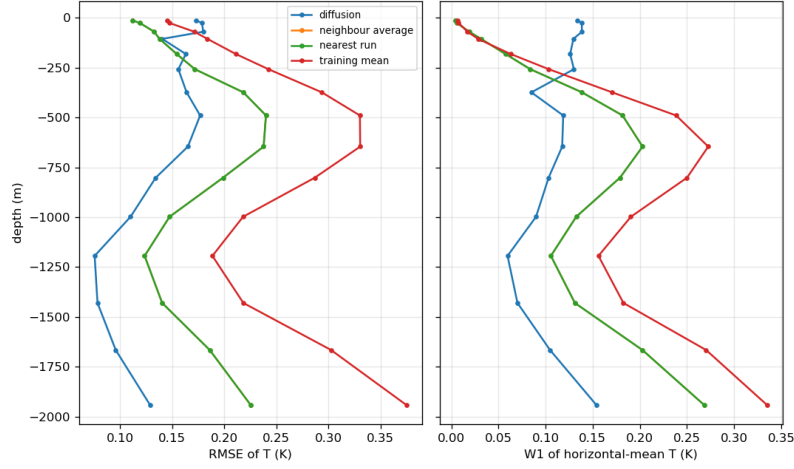


(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 9: Top row ($c_k = 0.8$ held out, extrapolation), 32 samples per run.



(a) Over the parameter grid: domain-mean T of the truth and of the generated states, their difference, and W_1 of the profile distribution. White dots: training runs; red crosses: held-out runs.



(b) RMSE (left) and W_1 (right) by depth, mean over the held-out runs.

Figure 10: Four training runs (one step in from each corner; 96 held out), 32 samples per run.

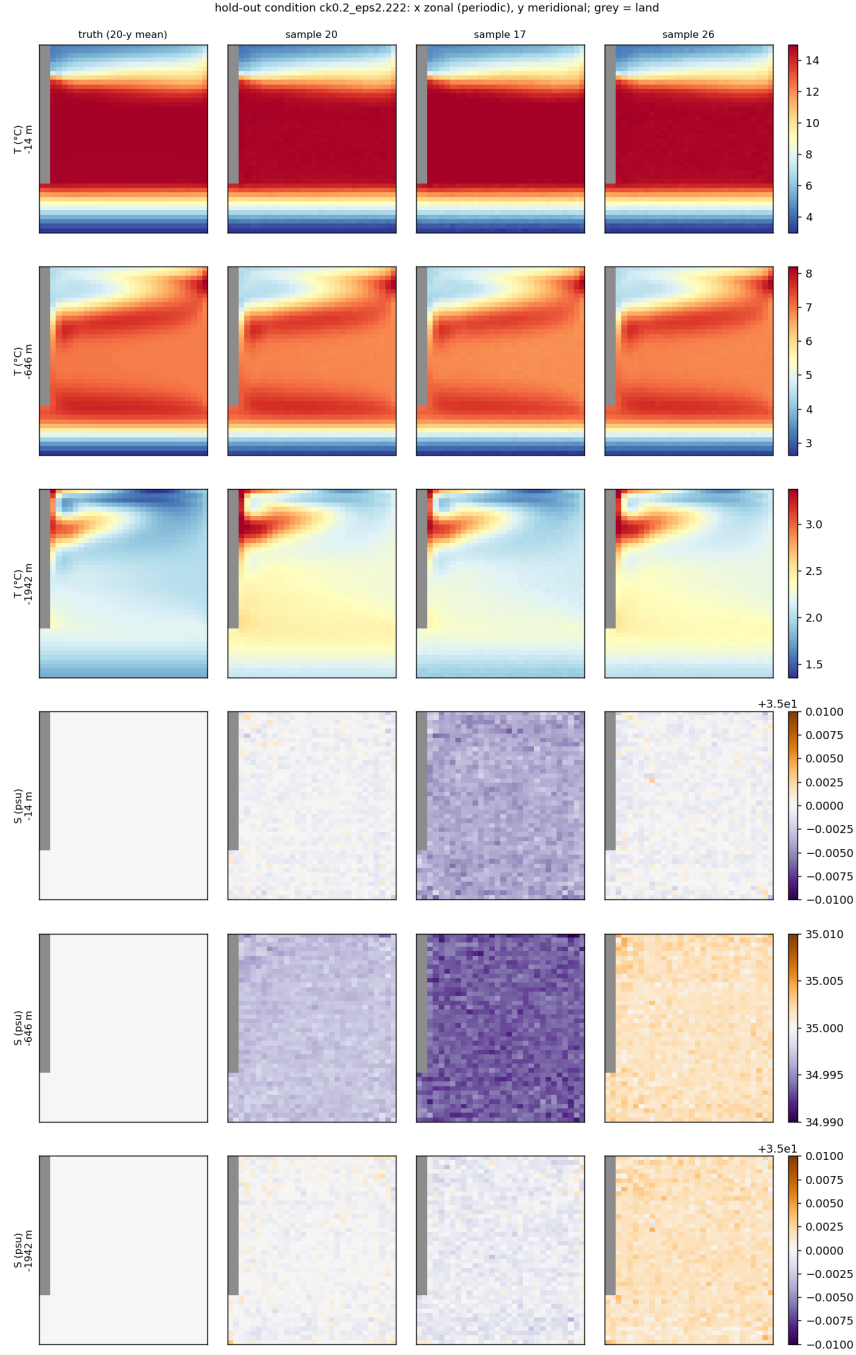


Figure 11: Scattered split, $c_k = 0.2$, $c_\epsilon = 2.222$: true 20-year mean (left) and three random samples at the surface, 650 m and the bottom; T (top three rows), S (bottom three rows). Grey: land.

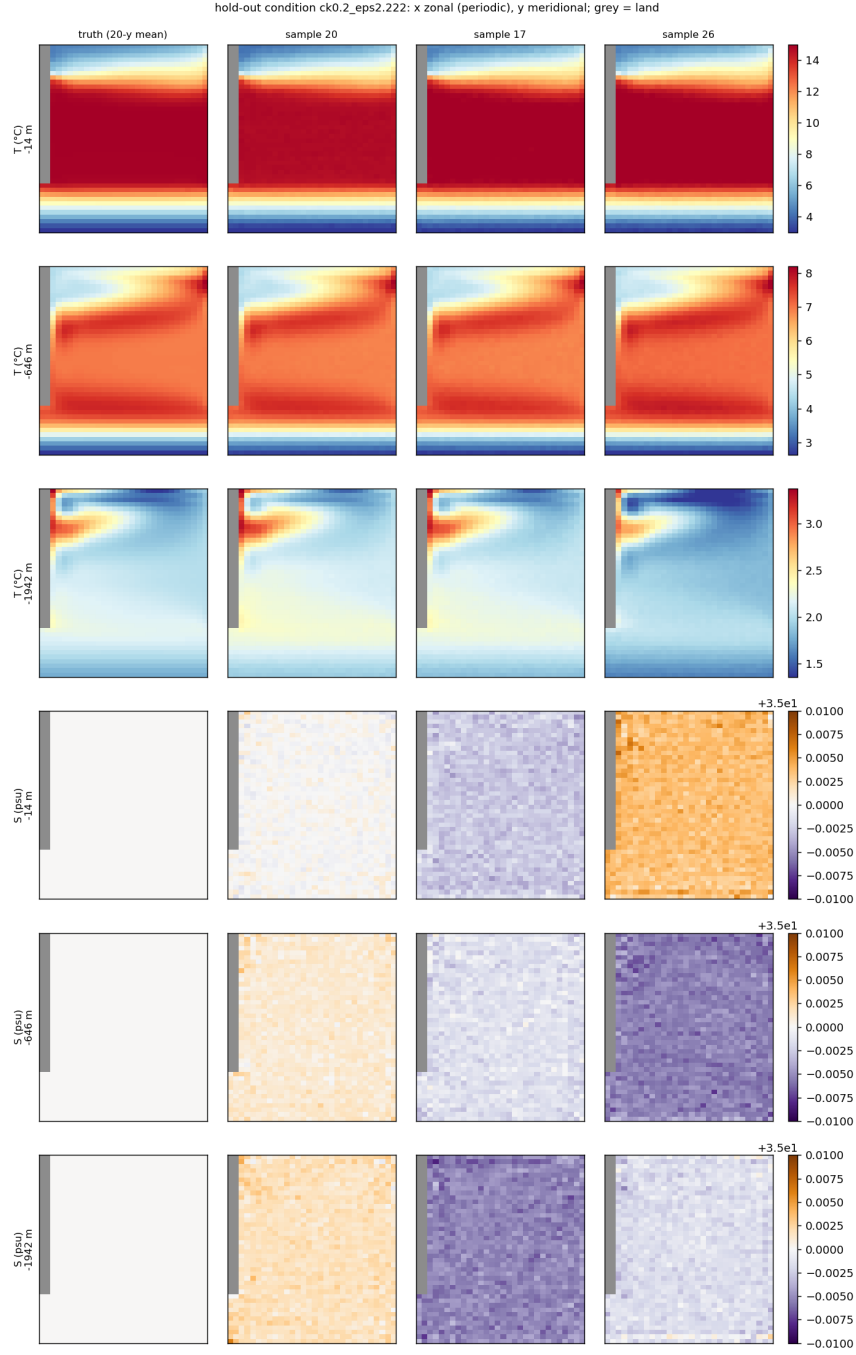


Figure 12: Band-of-three split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$, now in the middle held-out row).

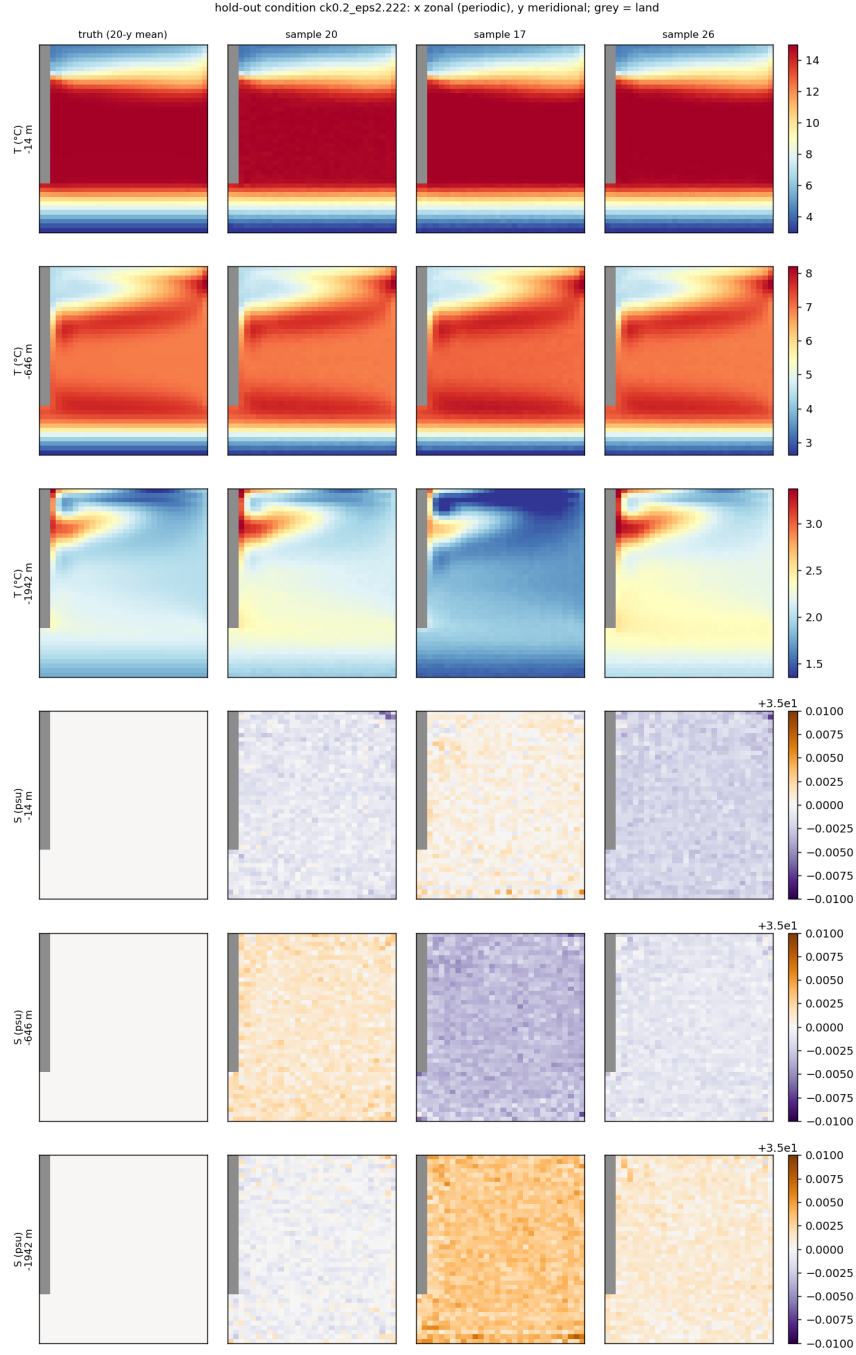


Figure 13: Centre- 5×5 split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$, inside the held-out block).

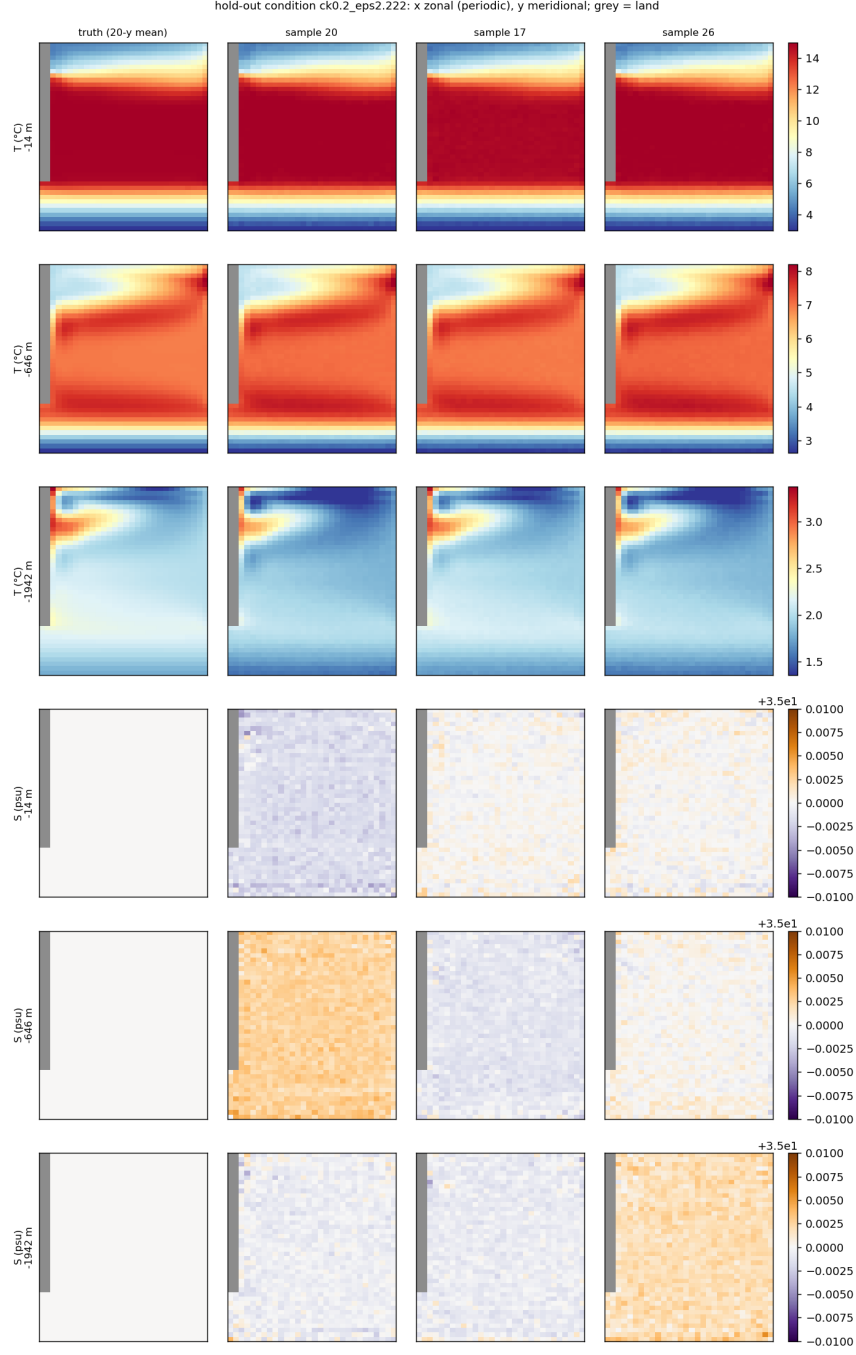


Figure 14: Centre- 7×7 split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$, inside the held-out block).

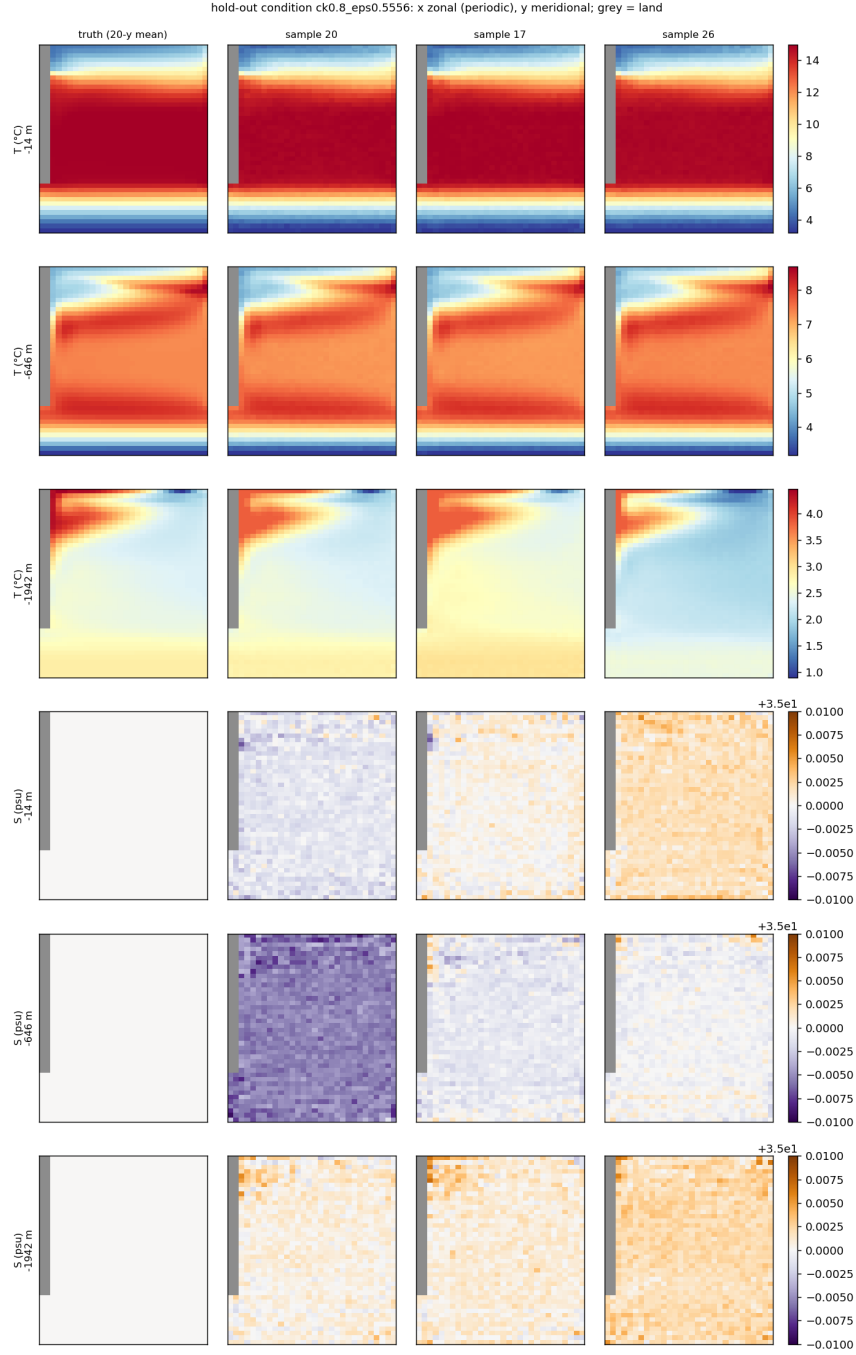


Figure 15: Outer-51 split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (same condition as the top-row split; here the whole outer ring is unseen).

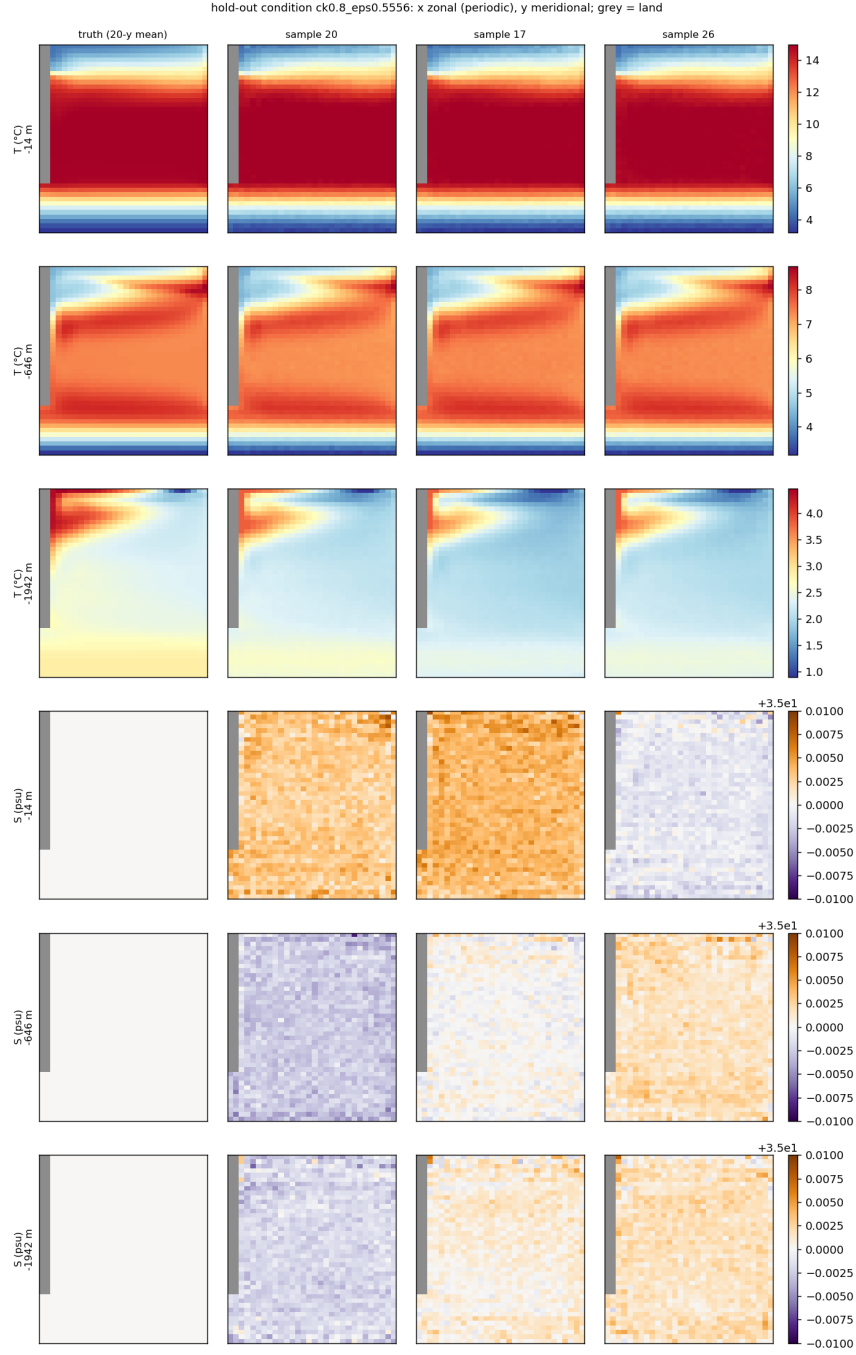


Figure 16: Outer-75 split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (training on the 5×5 centre only).

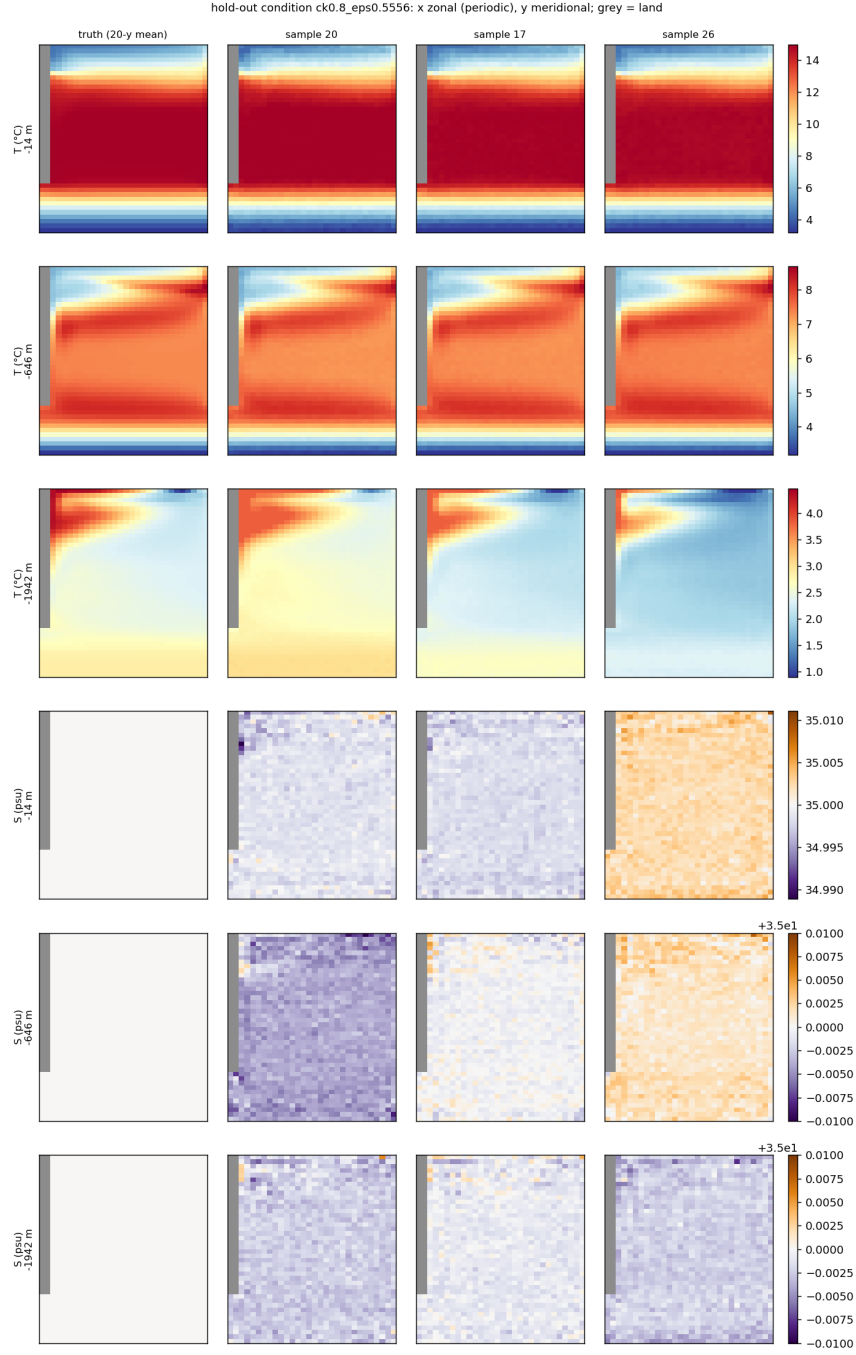


Figure 17: Top-row split, $c_k = 0.8$, $c_\epsilon = 0.5556$ (extrapolation one row above the training grid).

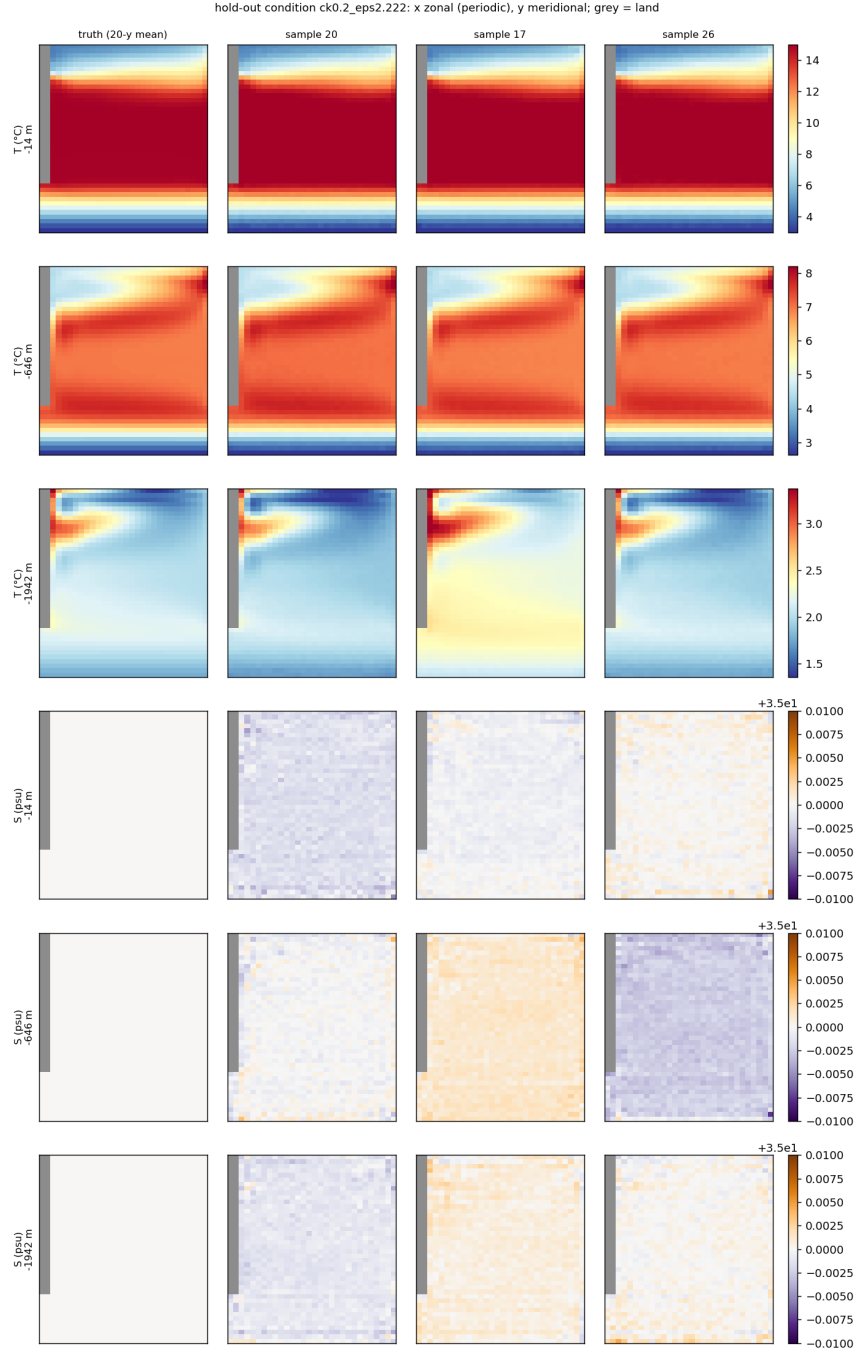


Figure 18: Four-run split, same condition ($c_k = 0.2$, $c_\epsilon = 2.222$).